

### PROFESSIONAL SUMMARY

- Over all 10+ years of professional IT experience and over 6 years in Data Engineering, and 4+ years in Datawarehouse.
- Experienced in working real time streaming with Kafka as data pipeline using spark streaming module.
- Experience with Apache Kafka and Azure Event Hubs for messaging and streaming applications using Scala.
- Optimized data ingestion, Data modelling, data Encryption and performance by tuning ETL workflows.
- Have a good working experience in Hadoop, Java, HDFS, Map-Reduce, Hive, Teg, Python, PySpark.
- Hands on working experience and developing large scale data pipelines using spark and hive
- Experience in using Apache Sqoop to import and export data to and from HDFS and Hive.
- Hands on experience in working with Azure Cloud and its components like Azure Data Factory, Azure Databricks, Logical Apps, Azure function Apps, snowflake and Azure DevOps services.
- Hands on working experience on working on Azure stack moving data from Data lake to Azure blob storage.
- Strong background in Data Load/Integration using the ADF
- Experience in building ETL (Azure Data Bricks) data pipelines leveraging PySpark, Spark SQL.
- Built and deployed scalable ETL pipelines using AWS Glue, PySpark, and Spark SQL to process large datasets and store results in optimized formats such as Parquet and ORC.
- Developed real-time data streaming solutions using Amazon Kinesis and AWS Lambda, enabling ingestion, processing, and storage of high-volume event data on Amazon S3.
- Designed and implemented data lakes and data warehouses using Amazon Redshift, AWS RDS, and S3 for structured and semi-structured data storage and analytics.
- Optimized data pipeline performance by tuning Spark configurations and Redshift queries using best practices like distribution styles, sort keys, and partitioning.
- Automated infrastructure provisioning and deployment pipelines using Terraform, AWS CloudFormation, and AWS Code Pipeline for CI/CD processes.
- Ensured data security and compliance by implementing AWS IAM policies, encryption using KMS, and logging/monitoring via AWS CloudWatch and AWS CloudTrail.
- Managed AWS Glue Crawlers and Catalogs for metadata management and seamless schema evolution in large-scale ETL workflows.
- Integrated legacy and cloud-native data systems using AWS Step Functions and Lambda to orchestrate event-driven ETL workflows with fault-tolerant design.
- Experience in developing pipelines in spark using Scala and Python.
- Experience working with Azure Logic APP Integration tool.
- Implemented azure functions, azure storage, and service bus queries for large enterprise level ERP integration system.
- Experience in creating and managing Azure DevOps tools for continuous integration and deployment (CI/CD) pipelines.
- Developed data ingestion workflows to read data from various sources and write it to Avro, Parquet, Sequence, JSON, and ORC file formats for efficient storage and retrieval.
- Hands-on experience in setting up workflow using Apache Oozie workflow engine for managing and scheduling Hadoop jobs.
- Experience in optimizing query performance in Hive using bucketing and partitioning techniques.
- Well versed in using ETL methodology for supporting corporate-wide- solution using Informatica.
- Optimized Spark jobs and workflows by tuning Spark configurations, partitioning and memory allocation settings.
- Extensive experience in develop, maintain and implementation of EDW, Data Marts, ODS and Data warehouse with Star schema and snowflake schema.
- Hands on experience with GitHub to push the code for maintaining versions.
- Comprehensive knowledge of Software Development Life Cycle and worked on Agile Methodology.

## EDUCATION

- Bachelors from JNTUH University, India.

## TECHNICAL SKILLS

|                                    |                                                                                                                                                                                           |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Big Data Technologies              | MapReduce, Hive, Teg, Python, Star Schema, PySpark, Scala, Kafka, Spark streaming, Oozie, Sqoop, Zookeeper, Databricks, Apache spark                                                      |
| Hadoop Distribution                | Cloudera, Horton Works                                                                                                                                                                    |
| Cloud Technologies                 | Azure data Factory, Azure Data Bricks, Logic Apps, Functional App, Snowflake, Azure DevOps, AWS (S3, Glue, Redshift, Data Lakehouse, Lambda, KMS, IAM Policies, Kinesis, EC2, CloudWatch) |
| Languages                          | Java, SQL, PL/SQL, Python, HiveQL, Scala.                                                                                                                                                 |
| Web Technologies                   | HTML, CSS, JavaScript, XML, JSP, Restful, SOAP                                                                                                                                            |
| Operating Systems                  | Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS.                                                                                                                                         |
| Build Automation tools & ETL Tools | Ant, Maven, Talend, Informatica, SSIS                                                                                                                                                     |
| Version Control                    | GIT, GitHub, Bitbucket                                                                                                                                                                    |
| Methodology                        | Agile, Scrum.                                                                                                                                                                             |
| IDE & Build Tools, Design          | Eclipse, Visual Studio.                                                                                                                                                                   |
| Databases                          | MS SQL Server 2016/2014/2012, Azure SQL DB, Azure Synapse. MS Excel, MS Access, Oracle 11g/12c, Snowflake, Cosmos DB                                                                      |

## PROFESSIONAL EXPERIENCE

**Role: Sr. Data Engineer**

**Aug 2024 – Till Now**

**Client: Genentech, San Francisco, CA**

### **Responsibilities:**

- Managed end-to-end Big Data flow within the application, including data ingestion from upstream sources to HDFS, as well as processing and analysis of data in HDFS.
- Analyzed data from Azure data storages using Databricks and Spark cluster capabilities, extracting valuable insights from large datasets.
- Perform all phases of software engineering including requirements analysis, application
- Developed and maintained end-to-end operations of ETL data pipeline and worked with large data sets in azure data factory.
- Increased the efficiency of data fetching by using queries for optimizing and indexing.
- Developed custom activities using Azure Functions, Azure Databricks, and PowerShell scripts to perform data transformations, data cleaning, and data validation.
- Hands on experience in using Kafka, Spark streaming, to process the streaming data in specific use cases.
- Implemented Azure EventHub for real-time data ingestion, enabling efficient streaming and processing of high-volume data.
- Containerized data processing applications using Docker, ensuring portability, scalability, and easier deployment across different environments.
- Designed and implemented Spark Streaming solutions for real-time data processing and analysis, enabling immediate insights and actionable intelligence.
- Utilized Terraform to automate infrastructure provisioning and management, ensuring consistent and reproducible deployments in an Azure environment.
- Configured data pipeline orchestration using YAML pipelines in Azure DevOps, ensuring efficient and reliable execution of data workflows.
- Conducted data validation, cleansing, and transformation processes, leveraging the power of SSIS/SSRS packages and SQL scripts to ensure data accuracy and integrity.

- Wrote SQL queries using programs such as DDL, DML and indexes, triggers, views, stored procedures, functions and packages.
- Worked closely with the data engineering team to enhance and optimize data pipelines, improving data processing speed and efficiency.
- Actively participated in code reviews, troubleshooting, and performance tuning sessions to improve overall system performance and reliability.
- Worked with Azure Logic Apps administrators to monitor and troubleshoot issues related to process automation and data processing pipelines.
- Developed CI/CD framework for data pipelines using Jenkins tool.
- Collaborated with DevOps engineers to developed automated CI/CD and test-driven development pipeline using azure as per the client requirement.
- Created and maintained HiveQL scripts and jobs using tools such as Apache Oozie and Apache Airflow.
- Working with JIRA to report on Projects, and creating sub tasks for Development, QA, and Partner validation.
- Experience in full breadth of Agile ceremonies, from daily stand-ups to internationally coordinated PI Planning

**Environment:** Azure Databricks, Data Factory, Logic Apps, Azure EventHub, containerization, spark streaming, data pipeline, terraform, azure DevOps, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, Snowflake, PySpark, GIT, JIRA, Jenkins, Kafka, ADF Pipeline, Power Bi.

**Role: Sr. Data Engineer**

**Aug 2022 – July 2024**

**Client: Walgreen, Chicago IL**

**Responsibilities:**

- Used Amazon Kinesis to stream real-time sales transactions and inventory updates from POS systems into AWS.
- Configured AWS Glue Crawlers to collect and CatLog raw sales and inventory data from multiple sources into Amazon S3.
- Built automated AWS Glue Jobs for ETL processes, transforming raw data into structured formats for analysis.
- Used AWS Lambda to process and transform real-time data from Kinesis streams.
- Stored raw and transformed data in Amazon S3 for long-term storage and as a data lake.
- Leveraged Amazon Redshift for structured data warehousing to enable high-performance querying of sales and inventory data.
- Created interactive Amazon Quick Sight dashboards for visualizing key metrics such as sales trends, inventory levels, and product profitability.
- Collaborated with business analysts to gather requirements and translate them into scalable data solutions.
- Created stored procedures and complex SQL queries to support analytical reporting.
- Supported integration of legacy on-premise systems with Snowflake cloud platform.
- Ensure that all data is encrypted at rest using S3 encryption and in transit using AWS KMS.
- Optimized S3 storage using S3 Lifecycle Policies to transition older data to more cost-efficient storage tiers like S3 Glacier.
- Implemented access control via AWS IAM to restrict access to sensitive genetic data.
- Automated the ETL processes using AWS Step Functions to orchestrate the entire data flow.
- Scheduled a regular data pipeline executions with AWS CloudWatch and AWS Lambda to ensure that the pipeline is updated with fresh data.
- Implemented Spark Streaming for real-time data processing, enabling quick insights and analytics on streaming data.
- Designed and implemented end-to-end data pipelines, ensuring efficient data flow, transformation, and integration across multiple systems.
- Utilized Terraform for infrastructure provisioning and automation, enabling seamless deployment and management of cloud resources.
- Implemented data ingestion and transformation processes for diverse data sources, ensuring data quality and integrity.
- Collaborated with cross-functional teams to gather requirements, design data models, and develop solutions that meet business needs.
- Performed data analysis and profiling to identify data patterns, anomalies, and insights for business decision-making.
- Implemented data governance and security measures to ensure compliance with industry regulations and protect sensitive data.

- Conducted performance tuning and optimization of data pipelines and queries to improve overall system efficiency.
- Provided technical guidance and support to team members, fostering knowledge sharing and collaboration.
- Wrote Hive queries for data analysis to meet the specified business requirements by creating Hive tables and working on them using Hive QL to simulate MapReduce functionalities.
- Created Partitions, Buckets based on State to further process using Bucket based Hive joins.
- Built and optimized large-scale ETL pipelines on Databricks using PySpark and SQL.
- Worked on RDD's & Data frames (Spark SQL) using PySpark for analysing and processing the data.
- Used Spark Streaming to divide streaming data into batches as an input to Spark engine for batch processing
- Implemented CICD pipelines to build and deploy the projects in Hadoop environment.
- Using JIRA to manage the issues/project workflow.
- Used Git as version control tools to maintain the code repository.

**Environment:** Amazon Kinesis, Real-time Data Streaming Glue, ETL (Extract, Transform, Load), Amazon S3, Data Lake Architecture, Amazon Redshift, Data Warehousing, SQL Analytics, Star Schema, Tibco Data Virtualization, Snowflake, Amazon Quick Sight, Business Intelligence (BI), AWS Lambda, Serverless Data Processing, AWS IAM (Identity and Access Management), AWS KMS (Key Management Service), Data Security, AWS CloudTrail, CloudWatch, Data Governance, Cost Optimization, spark streaming, data pipeline, terraform, Oracle, HDFS, MapReduce, Spark, Hive, SQL, Python, Snowflake, Scala, PySpark, GIT, JIRA, Jenkins, Kafka, Power Bi.

**Role: Big Data Engineer**

**June 2020 – July 2022**

**Client: Bank of America, Plano, TX**

**Responsibilities:**

- Utilized Sqoop to import data from MySQL to Hadoop Distributed File System (HDFS) on a regular basis, ensuring seamless data integration.
- Performed aggregations on large volumes of data using Apache Spark and Scala, and stored the processed data in Hive warehouse for further analysis.
- Worked extensively with Data Lakes and big data ecosystems, including Hadoop, Spark, Hortonworks, and Cloudera, to leverage their capabilities for efficient data processing.
- Developed Hive queries to analyse data and meet specific business requirements, utilizing Hive Query Language (HiveQL) to simulate MapReduce functionalities.
- Built HBase tables by leveraging HBase integration with Hive on the Analytics Zone, facilitating efficient storage and retrieval of data.
- Applied Kafka and Spark Streaming to process streaming data in specific use cases, enabling real-time data analysis and insights generation.
- Designed and implemented a data pipeline using Kafka, Spark, and Hive, ensuring seamless data ingestion, transformation, and analysis.
- Migrated existing data from RDBMS (Oracle) to Hadoop using Sqoop, facilitating efficient data processing and leveraging Hadoop's capabilities.
- Developed custom scripts and tools using Oracle's PL/SQL language to automate data validation, cleansing, and transformation processes, ensuring data accuracy and quality.
- Implemented Continuous Integration and Continuous Deployment (CI/CD) pipelines to build and deploy projects in the Hadoop environment, ensuring streamlined development and deployment processes.
- Utilized JIRA to manage project workflows, track issues, and collaborate effectively with cross-functional teams.
- Worked with Spark using Python (PySpark) and Spark SQL for faster data testing and processing, enabling efficient data analysis and insights generation.
- Employed Spark Streaming to divide streaming data into batches as input to the Spark engine for batch processing, facilitating real-time data processing and analytics.

- Utilized Zookeeper for coordination, synchronization, and serialization of servers within clusters, ensuring efficient and reliable distributed data processing.
- Worked on Oozie workflow engine for job scheduling, enabling seamless execution and management of data processing workflows.
- Leveraged Git as a version control tool to maintain code repositories, ensuring efficient collaboration, version tracking, and code management.
- Collaborated closely with the team to identify and resolve JVM-related issues, ensuring smooth execution and performance optimization.

**Environment:** Sqoop, MYSQL, HDFS, Apache Spark Scala, Hive Hadoop, Cloudera, HBASE, Kafka, MapReduce, Zookeeper, Oozie, Data Pipelines, RDBMS, AWS, EC2, Python, PySpark, Ambari, JIRA.

**Role: Hadoop Developer**

**Jun 2018 – May 2020**

**Client: Union Bank, Dallas, TX**

**Responsibilities**

- Developed ETL jobs using Spark -Scala to migrate data from Oracle to new MySQL tables.
- Rigorously used Spark -Scala (RRD's, Data frames, Spark SQL) and Spark - Cassandra -Connector API's for various tasks (Data migration, Business report generation etc.)
- Developed Spark Streaming application for real time sales analytics
- Analysed the source data and handled efficiently by modifying the data types. Used excel sheet, flat files, CSV files to generated PowerBI ad-hoc reports
- Analysed the SQL scripts and designed the solution to implement using PySpark
- Extracted the data from other data sources into HDFS using Sqoop
- Handled importing of data from various data sources, performed transformations using Hive, MapReduce, loaded data into HDFS.
- Extracted the data from MySQL into HDFS using Sqoop
- Implemented automation for deployments by using YAML scripts for massive builds and releases
- Apache Hive, Apache Pig, HBase, Apache Spark, Zookeeper, Flume, Kafka and Sqoop.
- Implemented Data classification algorithms using MapReduce design patterns.
- Extensively worked on creating combiners, Partitioning, distributed cache to improve the performance of MapReduce jobs.
- Worked on GIT to maintain source code in Git and GitHub repositories

**Environment:** Hadoop, Hive, spark, PySpark, Sqoop, Spark SQL, Cassandra, YAML, ETL.

**Role: Data warehouse Developer**

**May 2016 – May 2018**

**Client: AIG, Jersey City, NJ**

**Responsibilities:**

- Working as SQL Server Analyst / Developer / DBA using SQL Server 2012, 2015, 2016.
- Created jobs, SQL Mail Agent, Alerts and schedule DTS/SSIS Packages.
- Manage and update the Erwin models - Logical/Physical Data Modelling for Consolidated Data Store (CDS), Actuarial Data Mart (ADM), and Reference DB according to the user requirements.
- Source Controlling, environment specific script deployment tracking using TFS
- Export the current Data Models into PDF out of Erwin and publish them on to SharePoint for various users.
- Developing, Administering, and Managing corresponding databases: Consolidated Data Store, Reference Database (Source for the Code/Values of the Legacy Source Systems), and Actuarial Data Mart
- Writing Triggers, Stored Procedures, Functions, Coding using Transact-SQL (TSQL), create and maintain Physical Structures.

- Deployment of Scripts in different environments according to Configuration Management, Playbook requirements Create / Manage Files/File group - Table/Index association Query Tuning, Performance Tuning.
- Defect tracking and closing by using Quality Centre Maintain Users / Roles / Permissions.

**Environment:** SQL Server 2008/2012 Enterprise Edition, SSRS, SSIS, T-SQL, Windows Server 2003, Performance Point Server 2007, Oracle 10g, visual Studio 2010.

**Role: Data Warehouse Developer**

**Feb 2014 – Apr 2016**

**Client: Foot Locker, Bradenton, FL**

**Responsibilities:**

- Developed complex stored procedures, efficient triggers, and necessary functions, along with creating indexes and indexed views to optimize performance in SQL Server.
- Extensive experience in monitoring and tuning SQL Server performance, employing best practices to ensure optimal database performance.
- Expertise in designing ETL data flows using SSIS, creating mappings and workflows for extracting data from SQL Server, as well as performing data migration and transformation from Access/Excel sheets using SQL Server SSIS.
- Proficient in dimensional data modelling for Data Mart design, identifying facts and dimensions, and developing fact tables and dimension tables using Slowly Changing Dimensions (SCD) techniques.
- Skilled in error and event handling techniques, such as precedence constraints, breakpoints, check points, and logging, ensuring reliable and robust ETL processes.
- Experienced in building cubes and dimensions with different architectures and data sources for business intelligence purposes, including writing MDX scripting.
- Proficient in developing SSAS cubes, implementing aggregations, defining KPIs (Key Performance Indicators), managing measures, partitioning cubes, and creating data mining models. Deploying and processing SSAS objects.
- Experience in creating ad hoc reports and reports with complex formulas, utilizing querying capabilities of the database for business intelligence purposes.
- Expertise in developing parameterized, chart, graph, linked, dashboard, scorecards, and drill-down/drill-through reports on SSAS cubes using SSRS (SQL Server Reporting Services).

**Environment:** MS SQL Server 2016, Visual Studio 2017/2019, SSIS, Share point, MS Access, Team Foundation server, Git.